UNIT-1

- Data Communications and Networking for Today’s Enterprises :
  - Forces behind the evolution of data communications and networking
  - Enterprise Applications
  - Technologies Used
  - Key Components
  - Five components of Data Communication
  - Different forms of Information : Text, Numbers, Images, Audio, Video.
- Types of Connections (Modes) : Simplex, Half Duplex, Full Duplex.
- Types of Connections (Points) : Point-to-Point and Multipoint.
- Types of Communication Methods : Unicast, Broadcast and Multicast.
- Types of Network Topologies.
- All about Protocol / Layered architecture : Need, Elements, Whats a Protocol - Components.
- TCP/IP (Model) Architecture : All diagrams and Layers :
  - Application Layer.
  - Transport Layer.
  - Network Layer.
  - Data Link Layer.
  - Physical Layer.
- OSI (Model) Architecture : All diagrams and Layers :
  - Physical Layer.
  - Data Link Layer.
  - Network Layer.
  - Transport Layer.
  - Session Layer.
  - Presentation Layer.
  - Application Layer.
- Data and Signals : Definitions and Types.
- Digital Signals (NUMERICALS) : No of bits/level, bit rate, bit length - FORMULAS.
- Transmission Impairments :
  - Attenuation : Formula.
  - Distortion.
  - Noise.
- Digital-to-Digital Conversion : Line Coding and Block Coding.
- Digital-to-Digital Encoding (Theory + Diagrams + Graphs):
  - Unipolar.
  - Polar :
    - RZ (Return-to-Zero).
    - NRZ (Non RZ) :
      - NRZ-L.
      - NRZ-I.
    - Biphase :
      - Manchester.
      - Differential Manchester.
- Analog-to-Digital Encoding (Theory + Diagrams) :
  - Pulse Amplitude Modulation (PAM).
  - Pulse Code Modulation (PCM).

UNIT-2

- Bandwidth :
  - Definition.
  - Factors Affecting Bandwidth.
  - Formula.
- Throughput :
  - Definition.
  - Factors Affecting Throughput.
  - Formula.
- Delay / Latency :
  - 4 Components : d_{proc} , d_{queue}, d_{trans}, d_{prop}.
- Bandwidth Efficiency : Formula + Use + Methods : Multiplexing, Compression, etc…
- Bit Rate and Baud Rate : Formula.
- Multiplexing : Definition and Types (Diagrams) :
  - FDM (Frequency Division Multiplexing).
  - TDM (Time Division Multiplexing).
  - SDM (Statistical Division Multiplexing).
- DLL in OSI (Diagram) :
  - Sub-Layers : LLC and MAC.
  - Function of DLL : Framing, Addressing, Error Control, Flow Control and Access Control.
- Link Layer Addressing :
  - Introduction + Diagrams.
  - Three Types :
    - Unicast Address.
    - Multicast Address.
    - Broadcast Address.
- ARP (Address Resolution Protocol) :
  - Introduction + Diagrams.
  - Working.
- Standard Ethernet :
  - Introduction
  - Characteristics : Frame Format, Preamble, SFD, DA, SA, Length / Type, Data and CRC. Frame Length.
  - Addressing : Unicast, Multicasting and Broadcasting.
- MAC (Media Access Control) :
  - Introduction.
  - Types : CSMA, CSMA/CD, CSMA/CA.
- WLAN (Wireless LAN) : Introduction, Features, Frame Format, Types of WLAN Addresses and Types :
  - BSS (Basic Service Set) : Independent BSS and Infrastructure BSS.
  - ESS (Extended Service Set).

- Framing :
  - Introduction.
  - Techniques :
    - Character Count.
    - Byte Stuffing.
    - Bit Stuffing.

UNIT-3

- INTERNetworking :
  - Introduction.
  - Principles : 9 Principles.
- Network Layer :
  - Introduction.
  - Features.
  - Services.
  - Devices Used
- Internet Protocol :
  - IPv4 : Basics, Classes, Datagram Header (Diagram).
  - Public VS Private IP.
  - Subnet Mask :
    - Network + Host Portion.
    - Numericals : Finding Network Address, Broadcast Address and No of Hosts.
- CIDR (Classless Inter-Domain Routing) :
  - What ?
  - Advantages.
  - Notation.
  - CIDR -> Subnet Mask Conversion.
  - Numericals : Finding No of Hosts, Subnet Mask and Network Range
- VPN (Virtual Private Network) :
  - What ?
  - Why ?
  - Diagram.
  - How it works ?
  - Uses and Advantages.
- IPSec (IP Security) :
  - What ?
  - Why ?
  - Components.
  - How it works ?
  - Diagrams explaining the IPSec scenario.
  - Advantages
- DHCP (Dynamic Host Configuration Protocol) :
  - What ?
  - Why ?
  - How it Works ? :
    - DHCPDISCOVER.
    - DHCPOFFER
    - DHCPREQUEST
    - DHCPACK.
  - Packet Fields
- RIP (Routing Information Protocol) :
  - What ?
  - Characteristics.
  - How it Works ?
  - Types of RIP.
- OSPF (Open Shortest Path First) :
  - What ?
  - Key Features.
  - Areas.
  - How it Works ?
  - Advantages.
- OSPF vs RIP

UNIT-4

- Transport Layer :
  - Introduction.
- Protocols :
  - TCP :
    - Features,
    - Header.
  - UDP :
    - Features,
    - Header.
- DNS :
  - Introduction.
  - Types : Generic, Country and Inverse.
- HTTP :
  - Request and Response.
  - 4 Sections.
  - Methods, Status Codes and Examples.
- Compressions :
  - Lossy and Lossless.
  - Pros and Cons of both.
- MPEG Audio Compression - Steps.
- MPEG Video Compression - Steps.
- SIP :
  - Features.
  - Working.
  - Components.
  - Advantages.
  - Disadvantages.
- VoIP :
  - Features.
  - Working.
  - Advantages
  - Disadvantages.

RED - MOST IMPORTANT TOPICS.